STAT 5293 Spring 2026 — Project Proposal

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1 Objectives and Research Questions

This project aims to systematically evaluate whether structure-aware retrieval methods can improve factual reliability and reasoning quality in long-document question answering tasks. Specifically, we conduct a comparative study across three paradigms: (1) **No RAG**, where the model answers questions using only pretrained knowledge without external retrieval; (2) **Vector RAG**, a standard retrieval-augmented generation pipeline based on document chunking, embeddings, and similarity-based top-k retrieval; and (3) **TOC-Based RAG**, a structure-aware approach that leverages a document's hierarchical organization to iteratively navigate and retrieve relevant sections.

We aim to investigate three core research questions. First, does explicit document structure outperform similarity-based retrieval for long-document QA tasks? Second, how do different retrieval strategies influence factual reliability and hallucination rates? Third, what trade-offs exist between retrieval effectiveness and computational cost?

This project addresses several limitations in current retrieval-augmented systems. Similarity-based retrieval often overlooks document structure and long-range dependencies across sections. Long-context LLMs also suffer from the "Lost in the Middle" phenomenon, where critical information in the middle of long inputs is underutilized. While graph-based retrieval systems can improve reasoning capability, they are computationally expensive and complex to implement. Our work explores whether TOC-based retrieval can serve as a practical middle-ground solution between simple chunk-based retrieval and more expensive graph-based approaches.

We expect to obtain quantitative insights into how different retrieval paradigms influence answer accuracy, hallucination rates, and citation reliability. The study will also clarify scenarios in which structure-aware retrieval provides advantages and analyze trade-offs between retrieval quality and computational cost.

Expected deliverables include:

- A fully reproducible experimental pipeline
- Benchmark evaluation results on long-document QA
- Ablation studies for retrieval design choices
- A final report and presentation
- An open-source implementation of the system

This project aligns closely with the course objectives by evaluating real-world GenAI system design choices and comparing alternative architectural paradigms. It emphasizes experimental rigor, reproducibility, and systematic evaluation, while addressing reliability and hallucination mitigation in LLM-based systems.

2 Feasibility

The project is feasible within the course timeline because it does not require training foundation models and instead focuses on system design, retrieval strategy comparison, and controlled evaluation. We use the Qasper benchmark, which provides long academic documents with annotated QA pairs and is therefore well-suited for structured retrieval experiments. Since academic papers naturally contain section headers such as *Introduction*, *Method*, and *Experiments*, they are especially suitable for constructing a simple table-of-contents tree without requiring costly document restructuring from scratch.

Timeline and Milestones

Week	Milestone
1	Dataset preparation, benchmark familiarization, and baseline experiment setup
2	Implementation of the No-RAG and Vector RAG baselines
3	Implementation of the TOC-based RAG system
4	Construction of the evaluation pipeline and metric definitions
5	Full experiments and ablation studies
6	Result analysis, visualization, and final report preparation

We will rely on commonly available tools and resources, including HuggingFace Datasets, FAISS, Python-based LLM orchestration tools, and API-accessible language models. The project is also practical because the dataset already includes QA annotations, reducing the burden of constructing evaluation data from scratch.

Potential Challenges and Mitigation Strategies

Challenge	Mitigation Strategy
High cost of constructing QA pairs from scratch	Use the existing Qasper benchmark with annotated QA pairs and evidence
Complexity of TOC construction	Use document section headers from academic papers to build a simple hierarchical tree
High API cost from iterative retrieval	Limit evaluation set size during development and cap TOC navigation depth / iteration steps
Potential unfairness across retrieval methods	Standardize question sets, document sets, and retrieval budgets across all methods

The scope of the project is clearly limited to long-document QA, retrieval strategy comparison, and system-level evaluation. We do not aim to train new foundation models, pretrain embeddings, or build a production-scale retrieval system. This narrower scope helps ensure that the project remains realistic and well-defined for the course.

3 Innovation and Relevance

The originality of this project lies in its framing and methodology. While prior work often studies vector retrieval and graph-based retrieval separately, our project positions TOC-based retrieval as

a middle-ground paradigm between chunk-based similarity matching and computationally intensive graph reasoning. Rather than proposing an entirely new foundation model, we aim to contribute a controlled comparison of retrieval philosophies: no retrieval, similarity-based retrieval, and explicit structure-aware retrieval.

This project is also well connected to current GenAI research trends. It relates directly to recent work on structure-aware retrieval such as hierarchical RAG and RAPTOR, graph-based reasoning systems such as GraphRAG, and long-context limitations such as the "Lost in the Middle" effect. It also addresses one of the most practical problems in modern LLM systems: reducing hallucination by improving grounding and evidence selection.

The potential impact of the project is both academic and practical. From a research perspective, it can help clarify whether explicit document structure offers measurable benefits over similarity-based retrieval for long-document QA. From an application perspective, the findings may inform the design of more reliable enterprise RAG systems for domains such as scientific literature, legal documents, and technical documentation.

Our methodology is novel in that it combines a unified comparison across three retrieval paradigms, hierarchical document navigation via TOC trees, controlled evaluation on long-document QA, and multi-dimensional measurement across accuracy, hallucination, citation correctness, and efficiency. This gives the project both practical relevance and a clear research-oriented contribution.

4 Evaluation Plan

We evaluate system performance across multiple dimensions. First, we measure **answer accuracy**, which captures whether the final response matches the expected answer. Second, we measure **hallucination rate**, which indicates whether the system generates unsupported claims. Third, we measure **citation accuracy**, which tests whether the cited evidence actually supports the generated answer. In addition, we evaluate **retrieval effectiveness** through metrics such as Recall@k or evidence hit rate. Finally, we track **efficiency** using token usage, latency, and the number of API calls.

The experimental design is based on controlled comparisons across the three retrieval settings: No RAG, Vector RAG, and TOC-Based RAG. All systems will be tested on the same document corpus and question set to ensure fairness. We also plan to conduct ablation studies on major design choices in the TOC-based system, such as navigation depth, stopping thresholds, and greedy section selection strategies.

5 Conclusion

This project investigates whether explicit structural reasoning can outperform similarity-based retrieval for long-document question answering. By comparing No-RAG, Vector RAG, and TOC-Based RAG under a unified benchmark and evaluation framework, we aim to provide both practical insights for RAG system design and a meaningful contribution to current research on retrieval quality, reliability, and grounded generation in LLM systems.